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$$I = \int x^3 e^{x^2} dx$$

$$x^2 = t$$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{1}{2} dt$$

$$I = \frac{1}{2} \int t e^t dt$$

$$= \frac{1}{2} \left(e^t t - \int e^t dt \right)$$

$$= \frac{1}{2} e^t t - \frac{1}{2} e^t + C$$

$$= \frac{1}{2} e^t (t - 1) + C$$

$$= \frac{1}{2} e^{x^2} (x^2 - 1) + C$$

References